Templates for the Performance Task

I. Test For Safety and Practicality

A. Safety
   1. Is it safe for other people or animals that are involved?
      i. Unacceptable risks include indigestion or physical contact with any potentially hazardous materials, and undue physical, psychological, or emotional stress, including invasion of privacy.
      ii. Live animals (in particular vertebrate animals- those with a backbone) should be housed, cared for, and observed in a safe and humane manner.
   2. Address safety issues in your project proposal and check where the experiment will be performed and what safety gear will be used.

B. Practicality
   1. Measurability of changes to variables
   2. Availability of materials, time, and equipment

II. Proposal Form

The question I plan to investigate in my experiment (please phrase as a question):		
Question Checklist	**Yes**	**No**
1. Your teacher may put some restrictions on projects. Have you met your teacher's requirements?		
2. Is the topic interesting enough to read about, then work on for the next couple months?		
3. Can you find at least three (3) sources of written information on the subject?		
4. Can you measure changes to the important factors (variables) using a number that represents a quantity such as a count, percentage, length, width, weight, voltage, velocity, energy, time, etc.? Or, just as good, are you measuring a factor (variable) that is simply present or not present? For example: • Lights ON in one trial, then lights OFF in another trial • Use fertilizer in one trial, then DON'T USE fertilizers in another trial		
5. Can you design a "fair test" to answer your question? In other words, can you change only one factor (variable) at a time, and control other factors that might influence your experiment, so that they do not interfere.		
6. Is your experiment safe to perform?		
7. Do you have all the materials and equipment you need for your project, or will you be able to obtain them quickly and at a very low cost?		
8. Do you have enough time to do your experiment more than once before the due date?		

*Property of STI

III. Background Research Plan Worksheet

1. What question are you going to try to answer with the experiment?
2. List the keywords and phrases from your question and the topic in general.
3. Now use the keywords to build some questions to guide your background research. Develop at least two or three from each "question word". Don't worry about whether you already know the answer to the question- you'll find the answers when you do your background research. And don't forget to "network" with knowledgeable adults who can help guide you towards good materials!
4. To analyze the results from experiments you might need to know down any step or task that requires a formula or equation. Don't find the actual equation when you do your background research.
5. List steps or tasks that may require a formula or equation.

Background Research Plan Worksheet

Question Word	Possible Questions (You can think of others)	Substitute your keywords (or variations of your keywords) for the blanks in the previous column. Write down the relevant questions and use them to guide your background research.
Why	Why does ___ happen? Why does ____ ____? Why _____?	
How	How does _ happen? How does ___ work? How does ___ detect ____? How does one measure ____? How do we use ___? How _____?	
Who	Who needs ____? Who discovered ____? Who invented ____? Who _____?	
What	What causes ____ to increase/ decrease? What is ___ made of? What are the characteristics of ____? What is the relationship between __ and ___? What do we use ___ for? What _____?	

Where	When does ___ occur? When was ____ discovered? When _____?	
Where	Where does ___ occur? Where does ____ get used? Where _____?	

IV. Bibliography Worksheet

Use the format below in listing down the type of source.

This source is a ☐ Book ☐ Magazine ☐ Website ☐ Other: _____		
Author's Last Name:	First Name:	Middle Initial:
Date Published:	Publication/ Website Title	
Title of Article: (periodicals, encyclopedias, websites)		
Place Published (books only):	Publisher (books only):	Editor (if applicable)
Edition (if applicable)	Volume Number: Periodicals or Encyclopedias	Page Number(s):
Website is a ☐ Company ☐ Organization ☐ Government ☐ Newspaper/ Magazine ☐ Other		

V. The Research Paper

A. The purpose of the research paper is to give you the information to understand why your experiments turns out the way they do. The research paper should include the following aspects.
   1. The history of similar experiments or inventions
   2. Definitions of all important words and concepts that describe the experiment
   3. Answers to all background research questions
   4. Mathematical formulas, if any, needed to describe the results of experiment

B. See to it that you:
   1. collect information in an organized and accurate way to make writing the paper easier;
   2. set a source equipment; and
   3. use credible sources by reviewing the tips for assessing source quality you have learned in your Media and Information Literacy (MIL) class.

C. Observe the following when writing a research paper.
   1. Review the concept about plagiarism.
   2. Follow source citation format (MLA, APA, or some other) to use. Have clear transitions between paragraphs, and source citations.

VI. Research Paper Checklist

- ☐ Have you defined all important terms?
- ☐ Have you clearly answered all your research questions?
- ☐ Does background research enable you to make a prediction of what will occur in your experiment?
- ☐ Will you have the knowledge to understand what causes the behavior you observe?
  Does your research incldue the following:
  - Currently accepted theories, facts, and data
  - Relevant mathematics/equations (if applicable)
  - Key discoveries and early researchers
- ☐ Have you used the proper capitalization and punctuation?
- ☐ Have you checked your grammar and spelling?

VII. Variables and Hypothesis Worksheet

Your Hypothesis *(Fill in the blanks with the appropriate information from your own experiment.)*
If (I do this) _____ _____ _____, then (this) _____ _____ _____will happen.

Variables *(Fill in the appropriate information from your own experiment)*		
Independent Variable *(What will you be changing in the experiment. Note: There should only be one item listed here)*	**Dependent Variable** *(What will you be measuring or observing)*	**Controlled Variable** *(What will you be keeping the same during the experiment)*

Test the hypothesis by doing an experiment. First, design an experimental procedure; second, safety review. Prepare a detailed lists of the materials that you will use in conducting the experiment.

VIII. Conducting an Experiment

1. Do at least three (3) runs of your experiment.
2. Keep laboratory notebooks, single repositories of experiment data, observations, and reflections.
3. Remember that a failed experiment is not necessarily a failure. There is still a great deal of learning that can come from understanding what went wrong.

IX. Communicating the results

A. Final Report

This is the compilation of everything that you have done. It pulls together all information already created, including the research paper.

Final Report Checklist

- ☐ Does your abstract include a short summary of the hypothesis, materials & procedures, results, and conclusion?
- ☐ Have you used the proper capitalization and punctuation?
- ☐ Have you checked your grammar and spelling?
- ☐ Does your final report include the following key sections:
  1. Title Page
  2. Abstract
  3. Table of Contents
  4. Questions, Variables, and Hypothesis
  5. Background Research (your Research Paper)
  6. Material List
  7. Experimental Procedure
  8. Data Analysis and Discussion (including data tables and graphs)
  9. Conclusions
  10. Acknowledgments
  11. Bibliography

B. Class Presentation

1. Discuss your project in front of the class.
2. Communicates your project to others and including the sections related to all steps

Reference:

chemed.org. (n.d.). *Teacher's Guide to Science Projects*. Retrieved from http://www.chemed.org/wp-content/uploads/teachers-guide-to-science-projects-science-buddies.pdf